

Analyzing decision algorithms and problems

Outline

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1. Proof Methods

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2. Analysis of Algorithms

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3. Analysis of Problems
 1. Decidability
 2. Tractability

Methods of Learning

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Most RL research fits into the scientific method.

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Proof Strategies

1. **Direct proof**
2. **Proof by induction**
3. **Proof by contradiction**
4. Proof by contrapositive
5. Proof by exhaustion

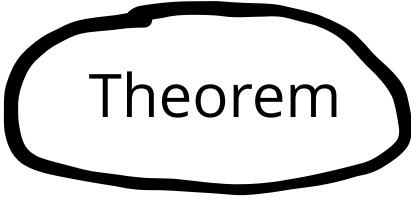
Direct Proof

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Definition

Direct Proof

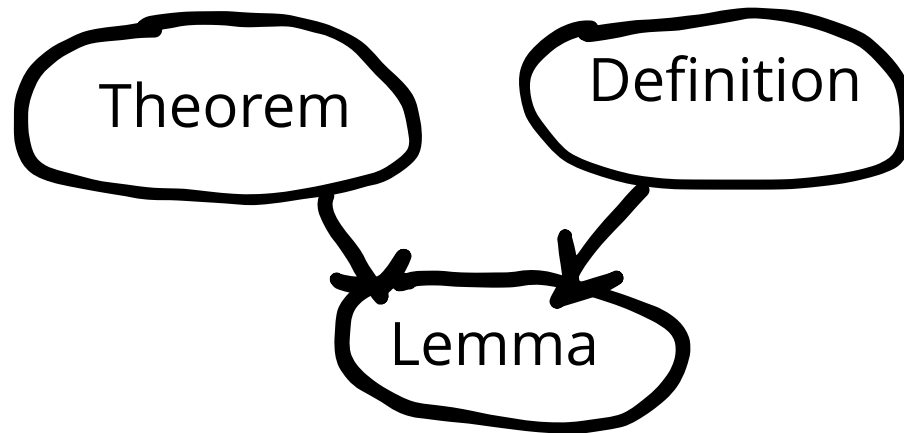


Theorem

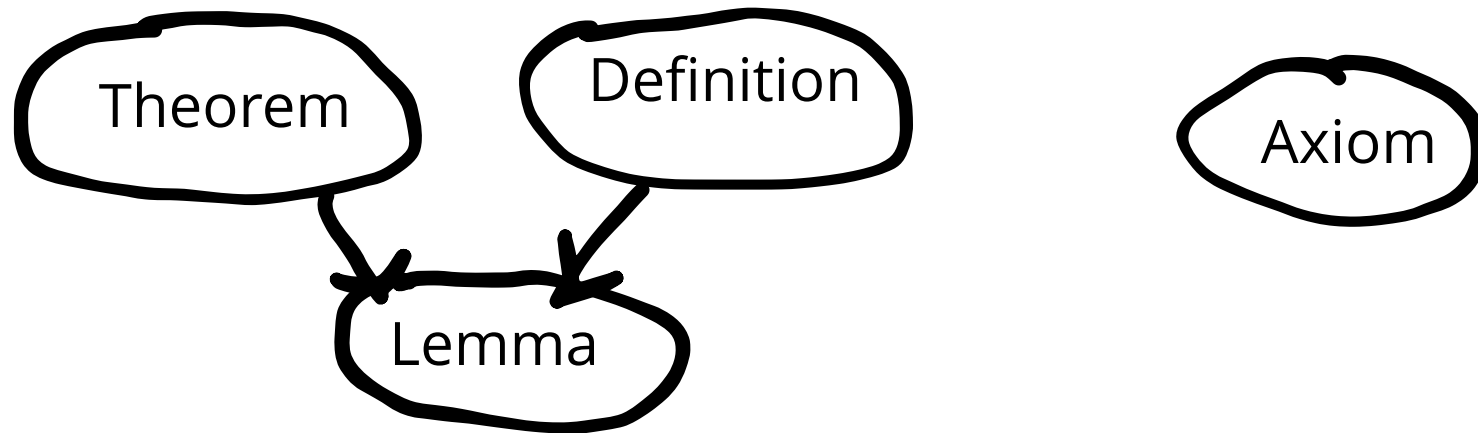


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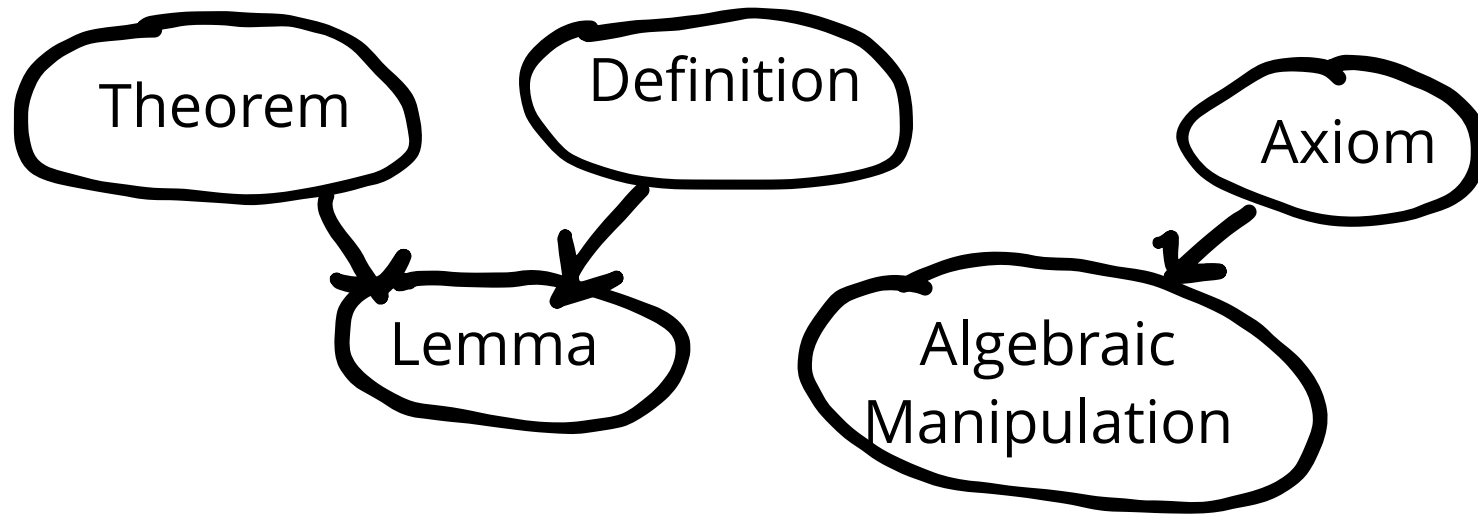
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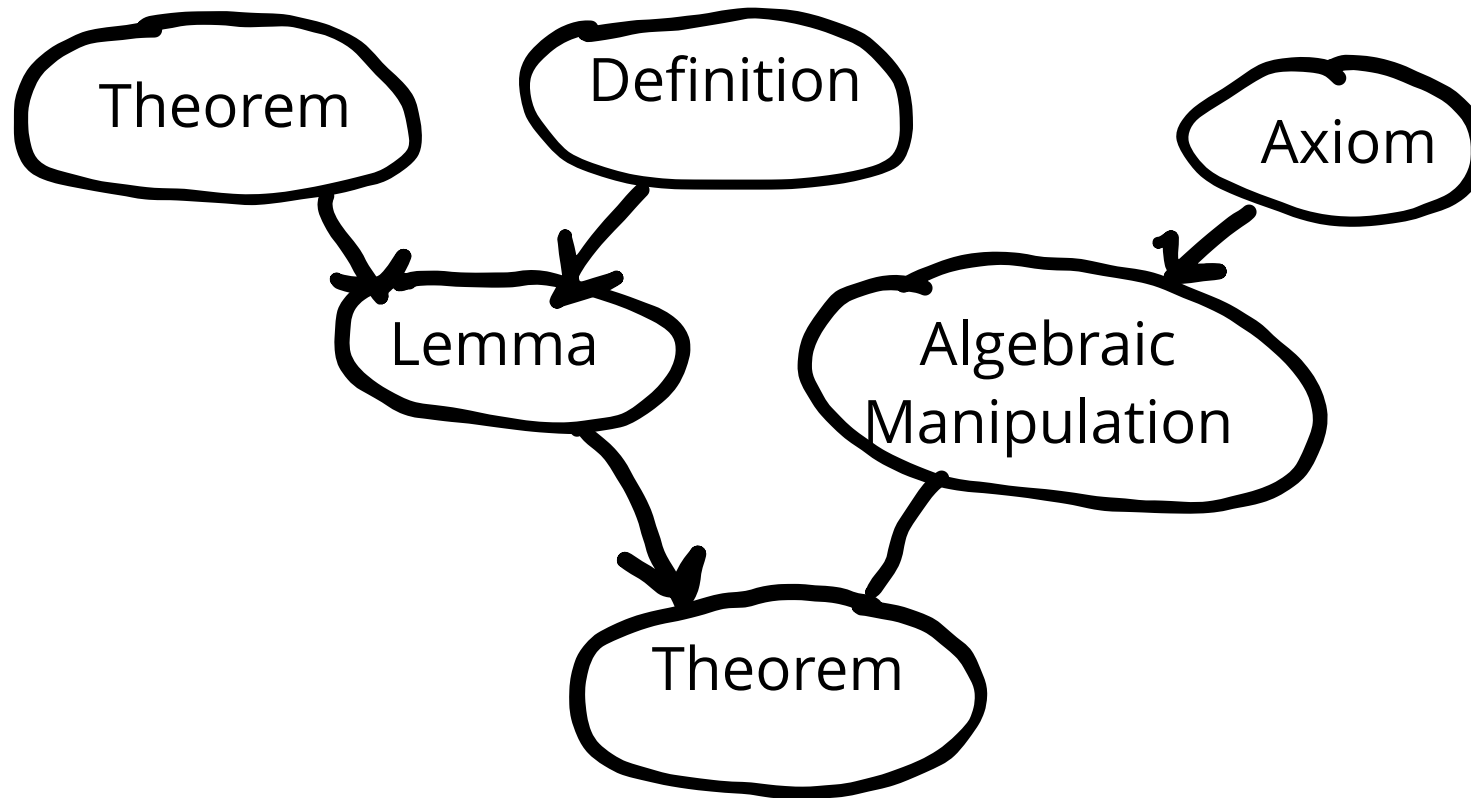
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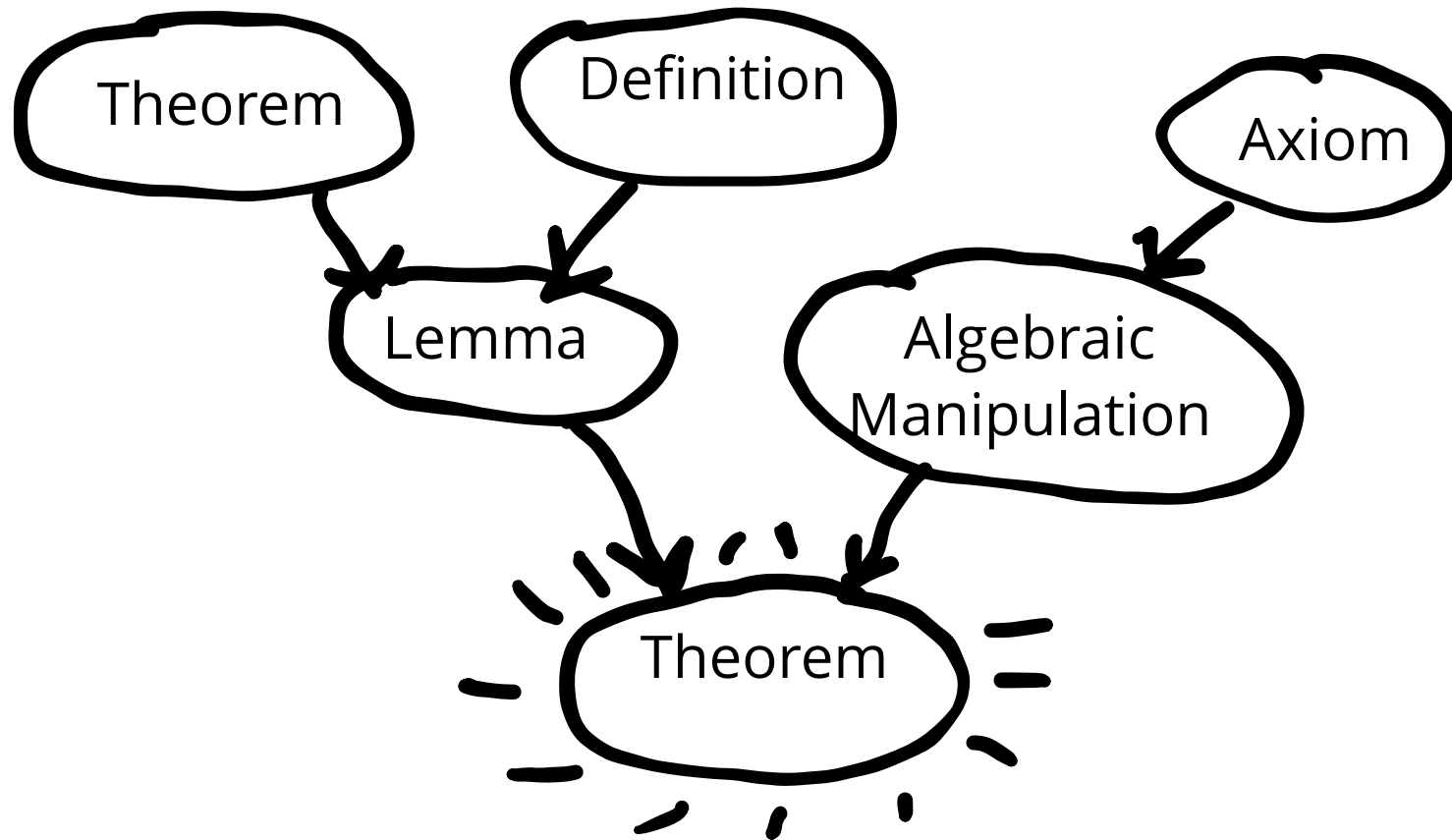
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Theorem: The sum of two even integers is even.

Proof: Let x and y be even integers and z be their sum. By definition, there exist integers a and b such that

$$x = 2a \text{ and}$$

$$y = 2b.$$

Then $z = x + y = 2a + 2b = 2(a + b)$. Since $a + b$ is an integer, z is even.

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1. Prove $P(1)$ (the base case)
2. Prove that $P(k) \implies P(k + 1)$ (the induction step)

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Proof by Induction Example

Theorem: For any natural number n , the sum of n and all natural numbers less than n is $\frac{n^2+n}{2}$

Proof: By induction:

- Base case: The sum of 1 and all natural numbers less than 1 is $1 = \frac{1^2+1}{2}$
- Induction step: If the sum of k and all natural numbers less than k is $\frac{k^2+k}{2}$, then the sum of $k+1$ and all natural numbers less than $k+1$ is $\frac{k^2+k}{2} + k + 1 =$

$$\frac{k^2+3k+2}{2} = \frac{(k+1)^2+k+1}{2}$$

$$\frac{(k+1)^2 + (k+1)}{2}$$

if $P(k)$ then $P(k+1)$

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1. Assume P is false, i.e. assume $\neg P$
2. Show that $\neg P$ implies two mutually contradictory assertions, Q and $\neg Q$
3. Q and $\neg Q$ cannot both be true (a "contradiction"), so P must be true

Proof by Contradiction

Example

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Theorem: $\sqrt{2}$ is irrational.

Proof by Contradiction

Example

Theorem: $\sqrt{2}$ is irrational.

Proof:

Assume $\neg P$

- Suppose $\sqrt{2}$ were rational. Then there exist positive integers a and b such that $\sqrt{2} = a/b$, **at least one of which is odd** (If a and b were both even, the fraction could be reduced). $\mathcal{Q}$
- Since $a^2 = 2b^2$, a^2 must be even, and thus a is even (since the square of an odd number is odd), and there exists some k such that $a = 2k$.
- Then $2b^2 = (2k)^2 = 4k^2$, so $b^2 = 2k^2$.
- Therefore b^2 is even, and b is also even. **Thus, both a and b are even.** This is a contradiction. $\neg \mathcal{Q}$

Algorithm Analysis

Computational Complexity: What is the number of basic operations an algorithm will use?

Asymptotic Notation

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$O(g(n)) = \{f(n) : \text{there exist positive constants } c \text{ and } n_0 \text{ such that}$
 $0 \leq f(n) \leq cg(n) \text{ for all } n \geq n_0\} .$

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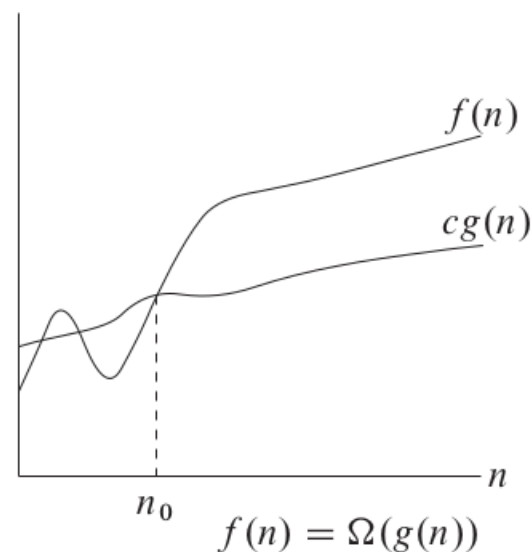
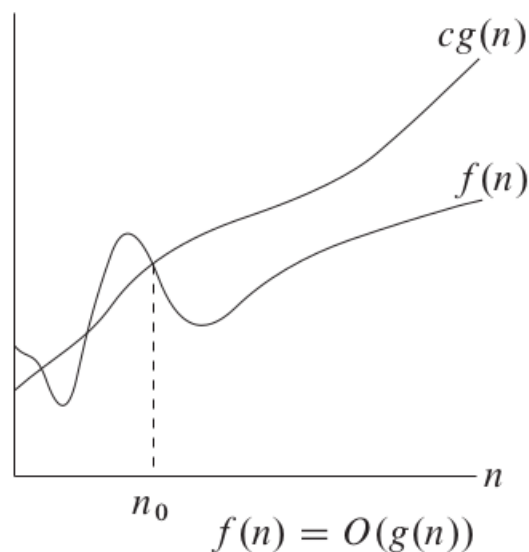
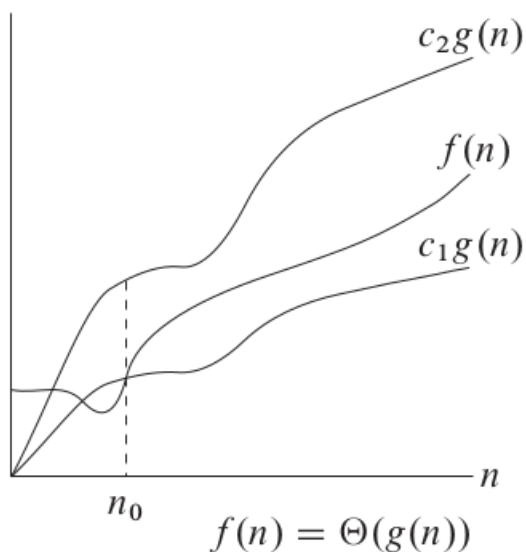
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Exercise: Complexity of Value Iteration for fixed horizon

Fixed Horizon Case:

for $t \in \{H - 1, \dots, 0\}$

for $s \in \mathcal{S}$

$$V_t^*(s) = \max_{a \in \mathcal{A}} (R(s, a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a) V_{t+1}^*(s'))$$

end

end

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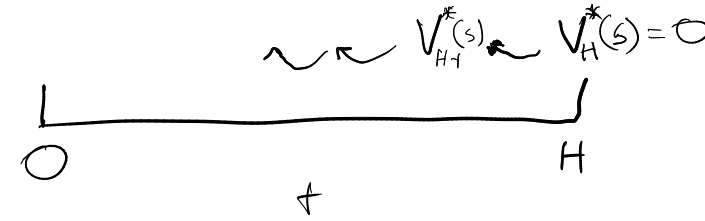
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end

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H

$|\mathcal{S}|$

$O(|\mathcal{A}||\mathcal{S}|)$

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$O(H|\mathcal{A}||\mathcal{S}|^2)$

Problem Analysis

Which problems can be solved and how hard are they?

Part 1: Decidability

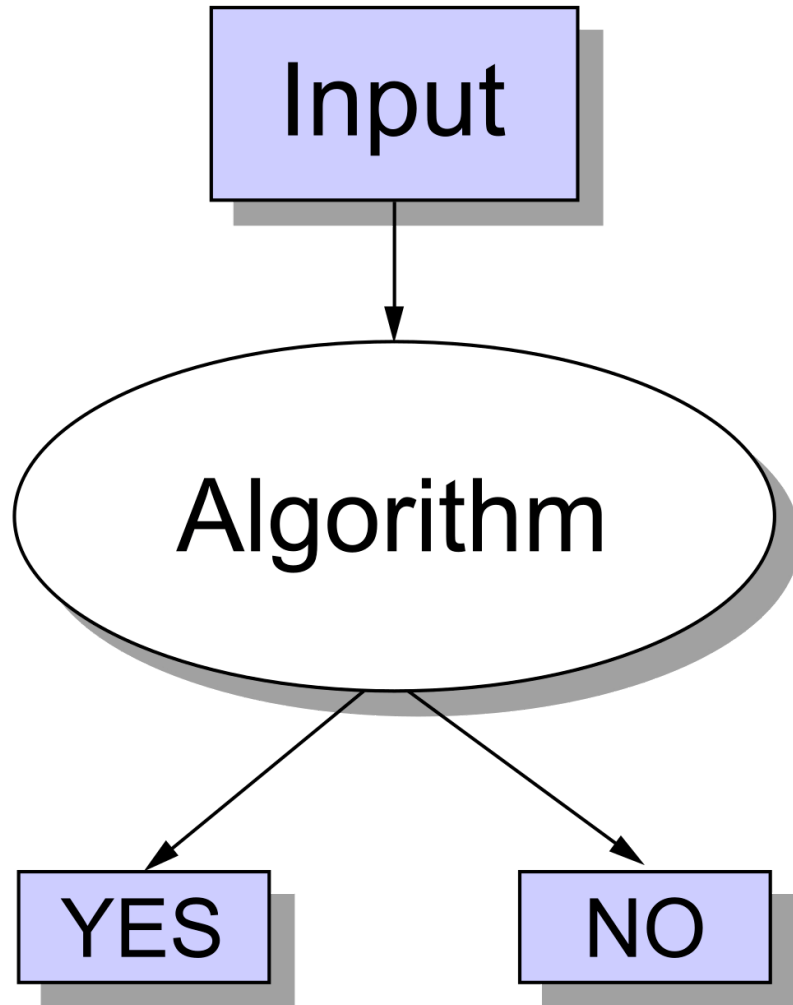
Decision Problems

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- A *Decision Problem* is a problem that has a yes-no answer for a set of inputs

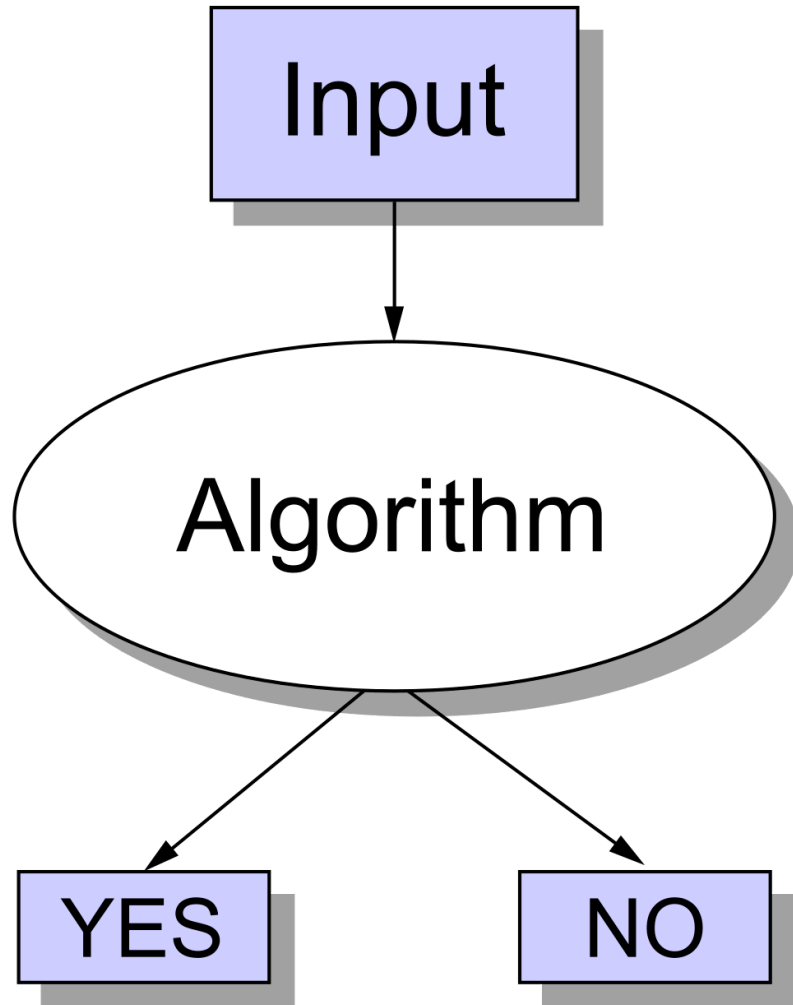
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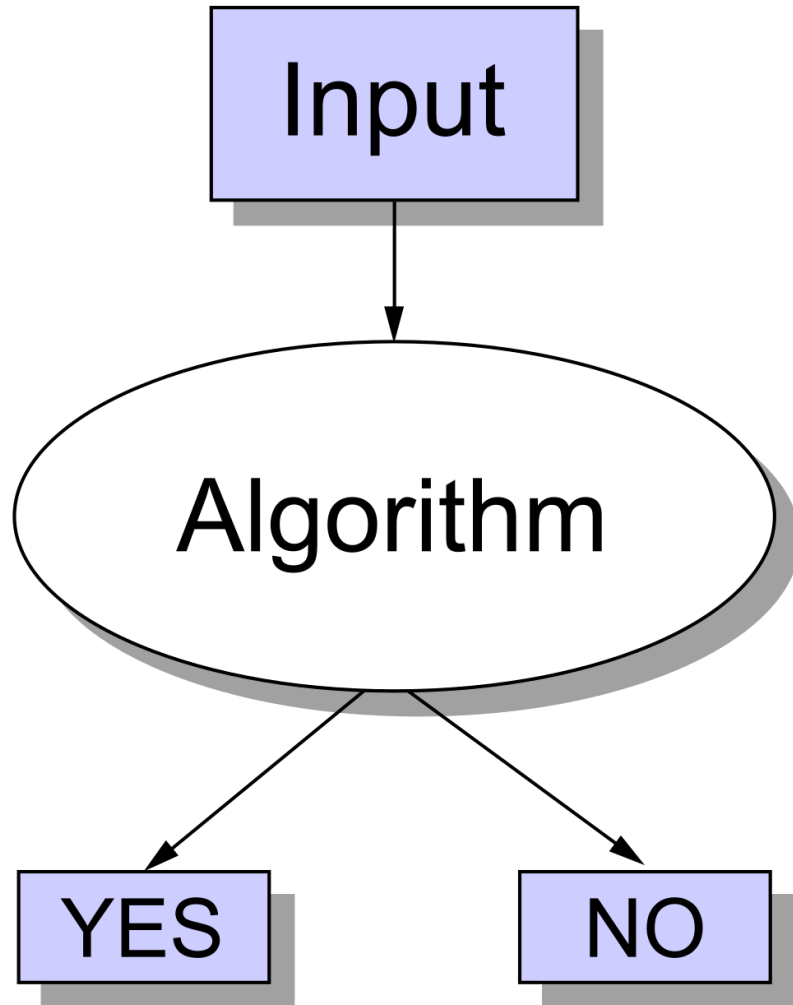


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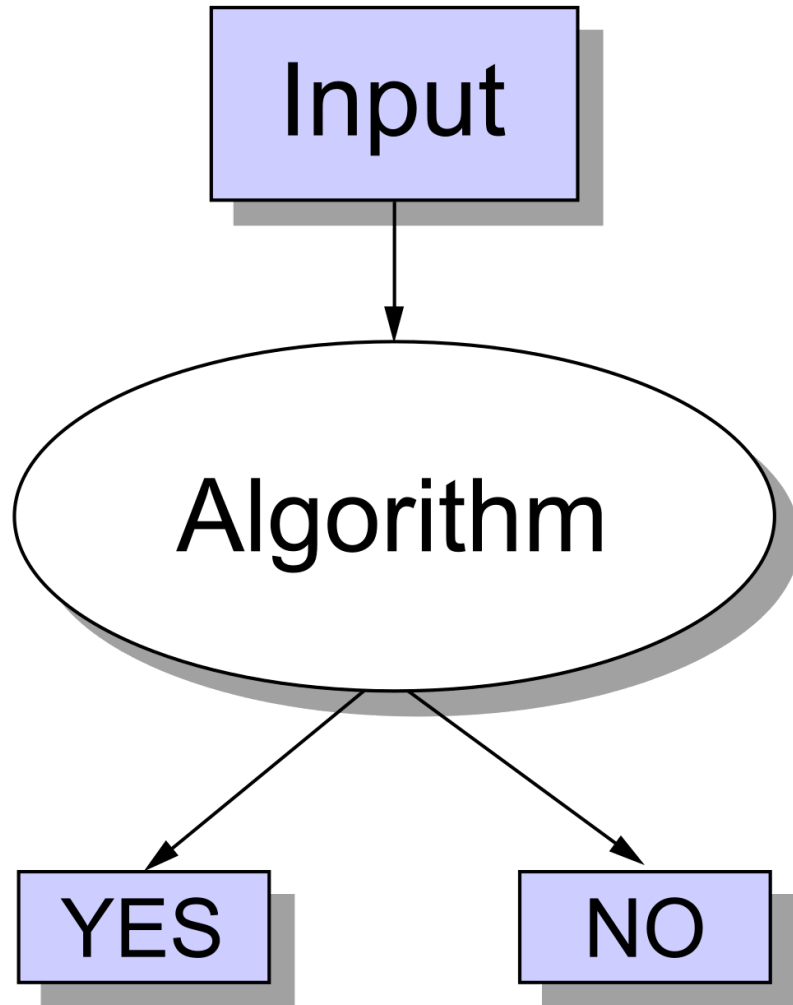


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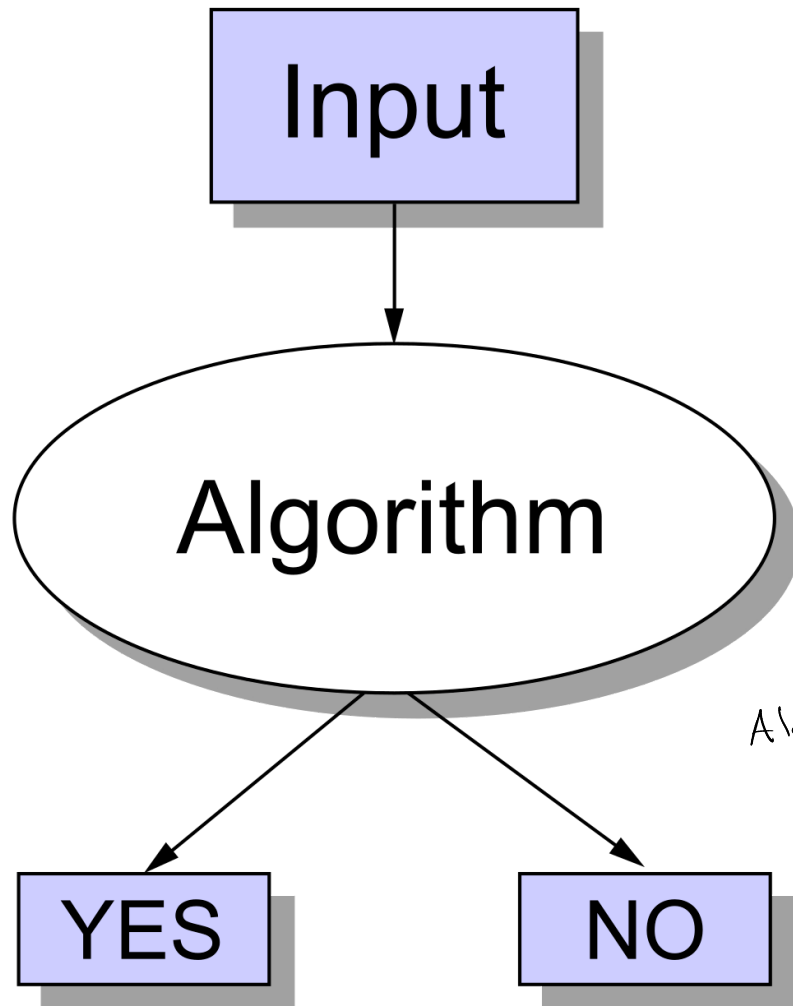
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Algo

A: Input: $S, A, R, T, \gamma, b_0, H$
Output: V^*

Algo B: Input: $S, A, R, T, \gamma, b_0, H, V$
Output: yes or no
1. Run A
2. Check if $V^* \geq V$

Finite-Horizon MDP Decision Problem:

- Inputs: $S, A, R, T, \gamma, b_0, H, V$
 S : initial state distribution
 A : action space
 R : reward function
 T : transition function
 γ : discount factor
 b_0 : initial state distribution
 H : horizon
 V : value bound
- Output: Is $E_{s \sim b_0} V_0^*(s) \geq \underline{V}$?

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Is the finite horizon MDP decision problem decidable?

Yes.

Algorithm:

Run finite horizon VI which terminates in $O(H|S|^2|A|)$ time
Check whether $V^* \geq \underline{V}$

Decidability

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How can we prove that the halting problem is undecidable?

Basic idea (this is NOT a rigorous proof): Proof by contradiction

- Assume $h(f)$ returns true iff f halts
- Define a function g that loops forever if $h(g)$ is true (doing this rigorously is more technical than we will get into)
- Then $h(g)$ cannot return true or false, a contradiction

Proof by Reduction

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We know the halting problem is undecidable. Can we use this to prove that other problems are undecidable?

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If the halting problem is reducible to B , then B is undecidable.

Part 2: Tractability

Tractability

Tractability

$$O(1)$$

$$O(\log(n))$$

$$O(n)$$

$$O(n^2)$$

$$O(n^k)$$

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Tractability

What definition should we use to separate tractable from intractable problems?

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Table 2.1 The running times (rounded up) of different algorithms on inputs of increasing size, for a processor performing a million high-level instructions per second. In cases where the running time exceeds 10^{25} years, we simply record the algorithm as taking a very long time.

	n	$n \log_2 n$	n^2	n^3	1.5^n	2^n	$n!$
$n = 10$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	4 sec
$n = 30$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	18 min	10^{25} years
$n = 50$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	11 min	36 years	very long
$n = 100$	< 1 sec	< 1 sec	< 1 sec	1 sec	12,892 years	10^{17} years	very long
$n = 1,000$	< 1 sec	< 1 sec	1 sec	18 min	very long	very long	very long
$n = 10,000$	< 1 sec	< 1 sec	2 min	12 days	very long	very long	very long
$n = 100,000$	< 1 sec	2 sec	3 hours	32 years	very long	very long	very long
$n = 1,000,000$	1 sec	20 sec	12 days	31,710 years	very long	very long	very long

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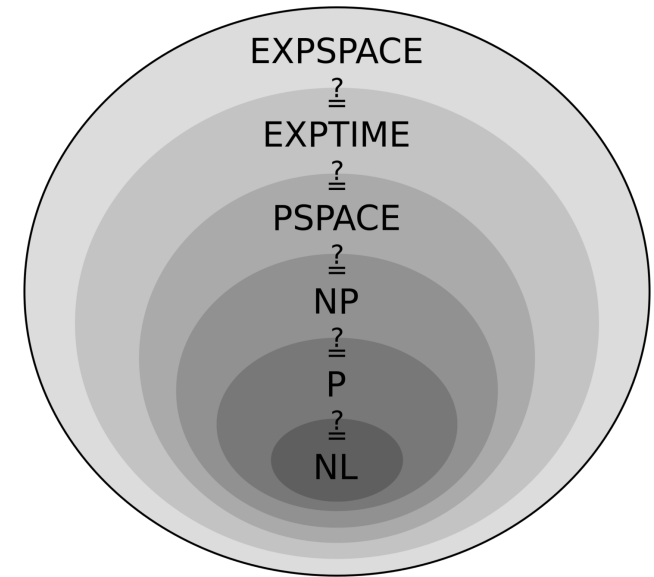
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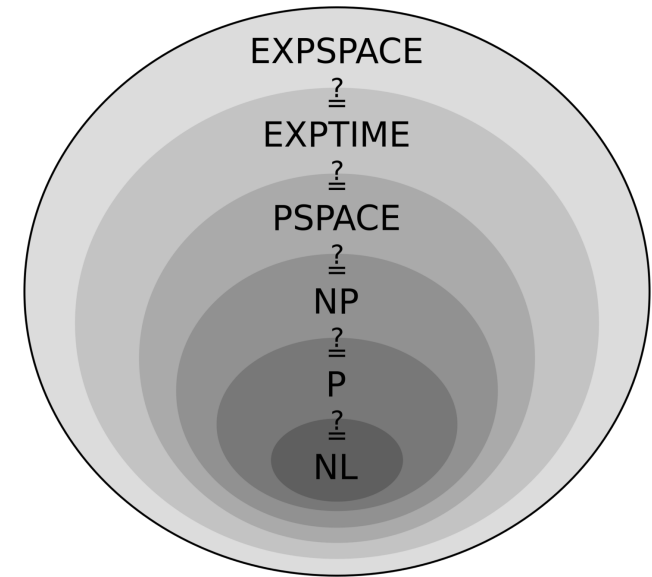
We say that a problem is *tractable* if it can be solved in $O(n^k)$ for some k .

Complexity Classes



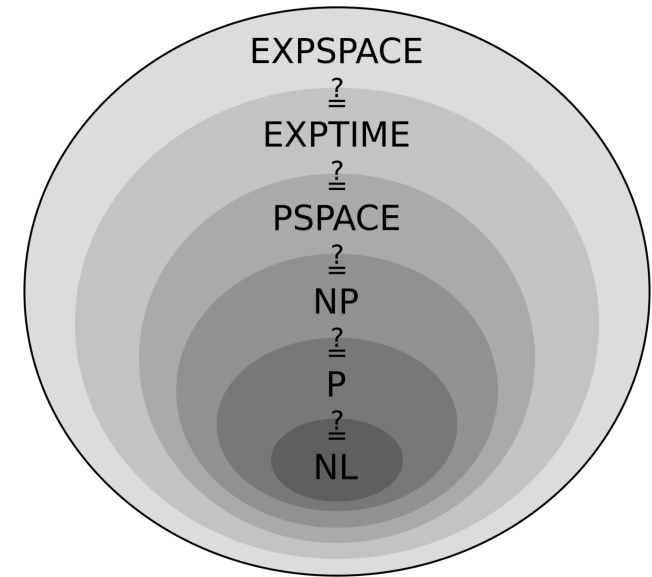
Complexity Classes

- P: Any instance can be solved in polynomial time in the size of the input
 - Is the Finite Horizon MDP a member of P?



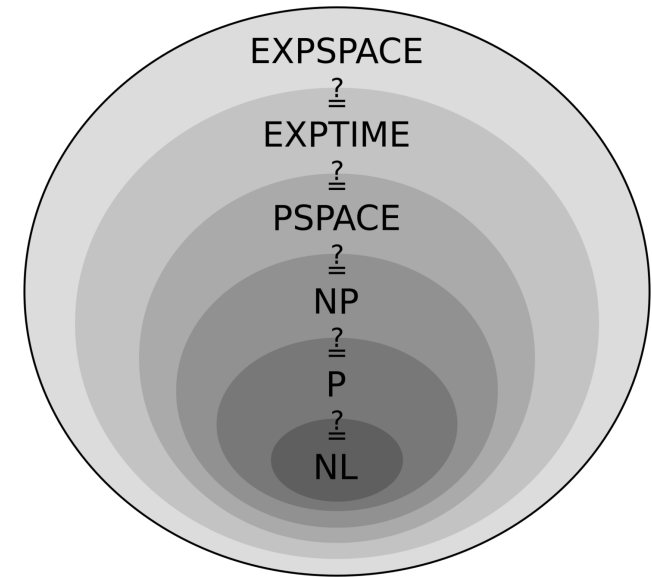
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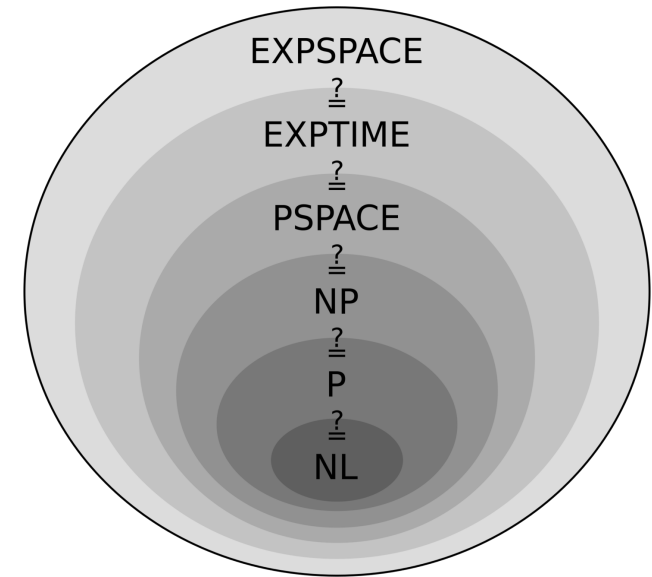
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How can we prove that a problem is a member of a class?

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How can we prove that a problem is a member of a class?

By finding an algorithm that solves the problem within the computational limit.

The NP Complexity Class

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- "Nondeterministic Polynomial"

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$$P \subset NP$$

(We think - this is one of the biggest
unsolved problems in CS)

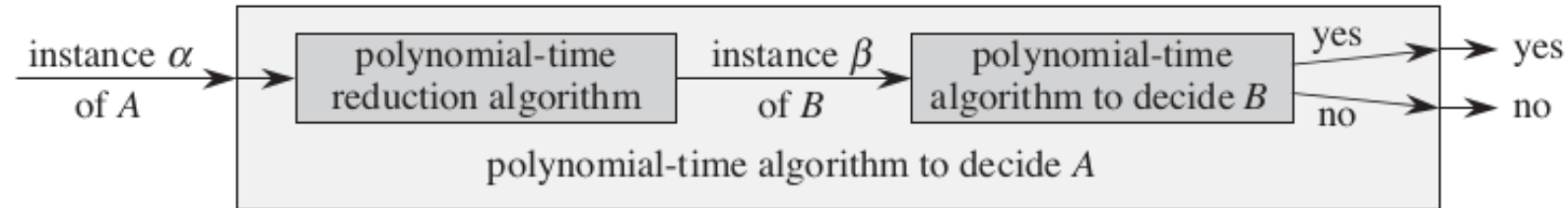
Polynomial Reduction

Polynomial Reduction

How can we prove that problem B is at least as hard as problem A?

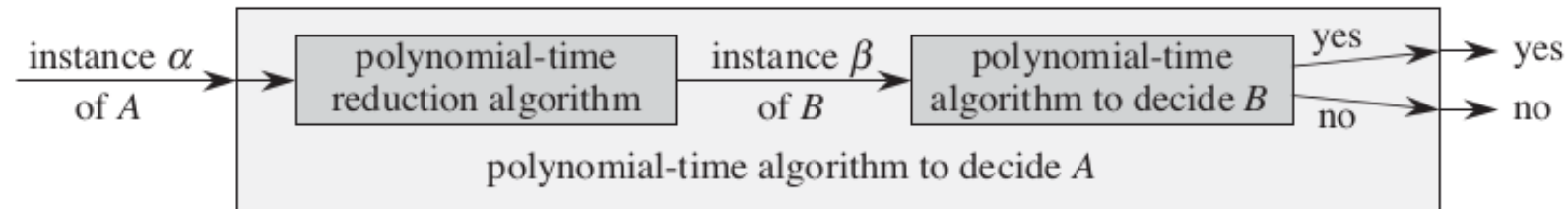
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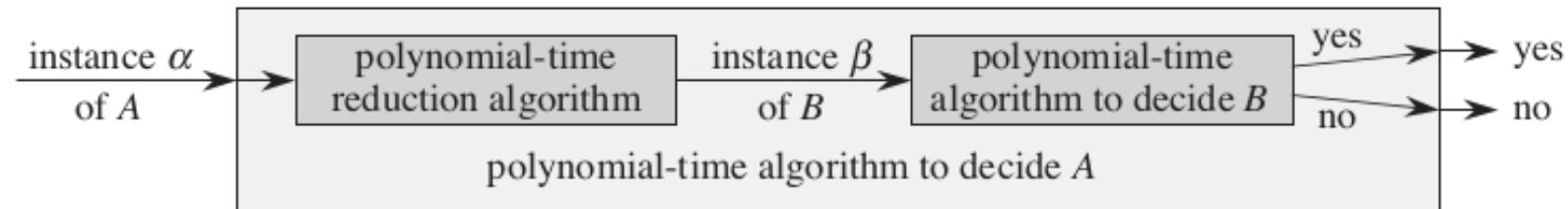
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$$A \leq_P B$$

A is no harder than B

NP Completeness

Example proving that a problem is NP-complete is in section 34.3 of "Introduction to Algorithms"

NP Completeness

Problem A is said to be *NP-complete* if $A \in \text{NP}$ and

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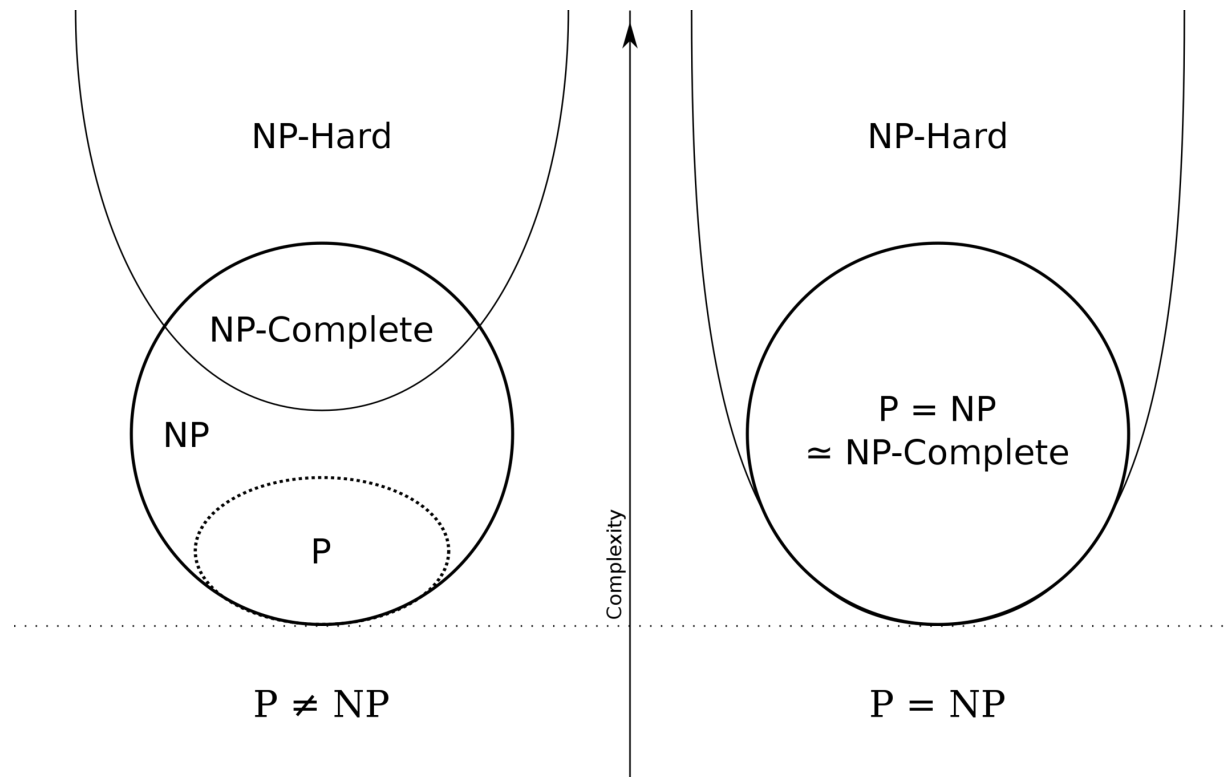
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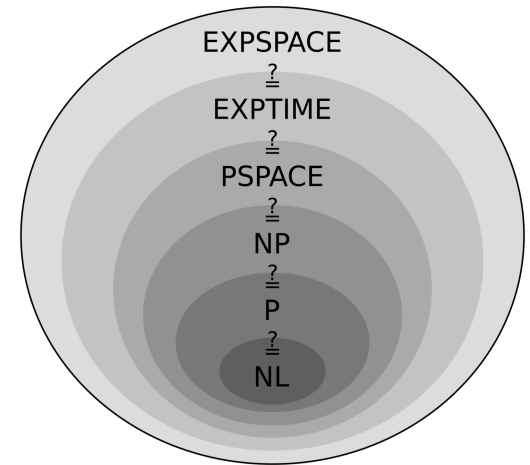
- 3-SAT is NP-complete (Cook, 1971)
- If 3-SAT can be reduced to $A \in \text{NP}$, A is also NP-complete!

NP-Hardness

- NP-hard: at least as hard as all of the problems in NP
- NP-complete: NP-hard and in NP

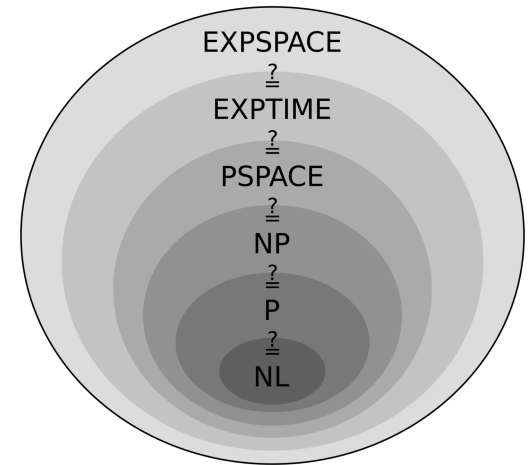


PSPACE Complexity Class



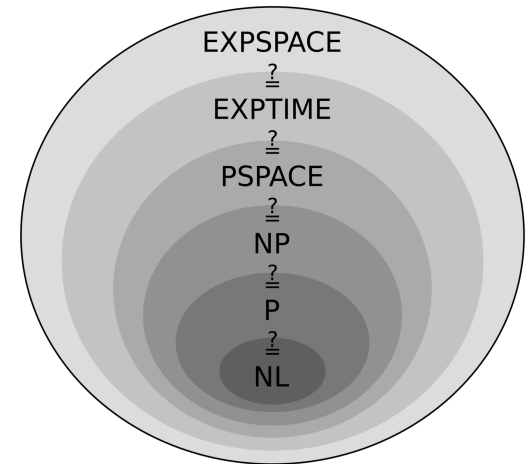
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- PSPACE: Set of all problems that can be solved using polynomial *space*



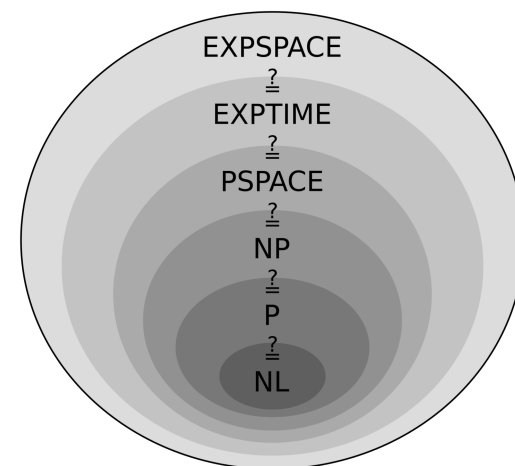
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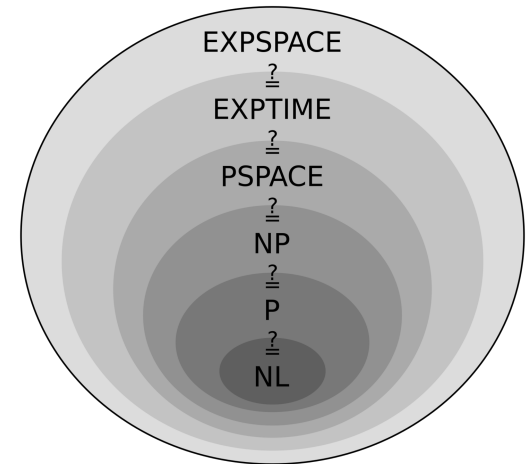
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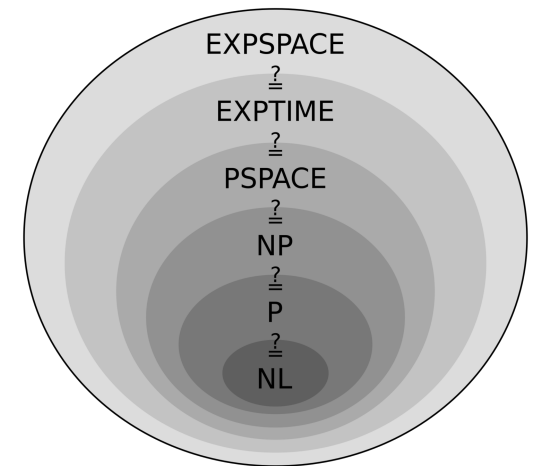
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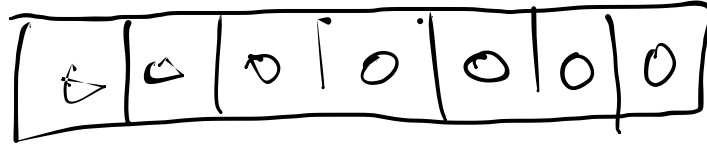
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- $NP \subset PSPACE$???





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- QSAT is PSPACE-complete (Stockmeyer and Meyer, 1973)
- If QSAT can be reduced to A, A is PSPACE-hard

Review

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1. Three most common proof methods:

1. Direct
2. Induction
3. Contradiction

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5. Tractability

1. Problem is tractable if there is a polynomial-time algorithm that decides it
2. Complexity Classes: P, NP, PSPACE, etc.
3. Can prove problems are in a class by providing an algorithm, complete by reducing a complete problem to it and proving it is in the class